MU-MIMO

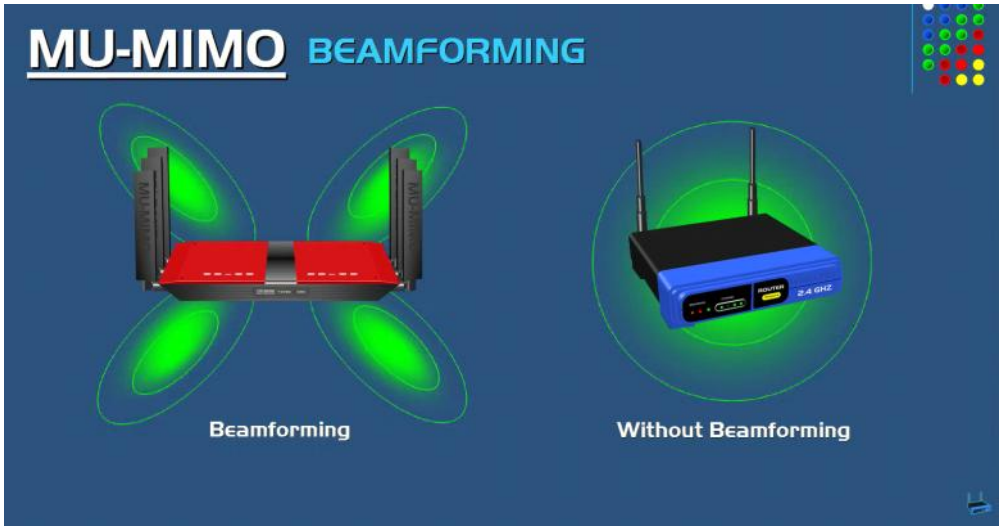
Monday, March 23, 2026 4:24 PM

MU-MIMO (Multi-User Multiple Input Multiple Output) is a Wi-Fi technology introduced in 802.11ac Wave 2 that allows a router to communicate with multiple devices at the same time.

Earlier, SU-MIMO could only serve one device at a time, making others wait, which caused lag when many devices were connected also it wor ks on first come first server basis

With MU-MIMO:

* Multiple devices receive data simultaneously → faster speeds and less buffering
* Best for high usage like streaming and downloads
* Works on 5 GHz band in Wi-Fi 5
* Supports up to 4 simultaneous streams (based on antennas like 2x2, 3x3, 4x4)

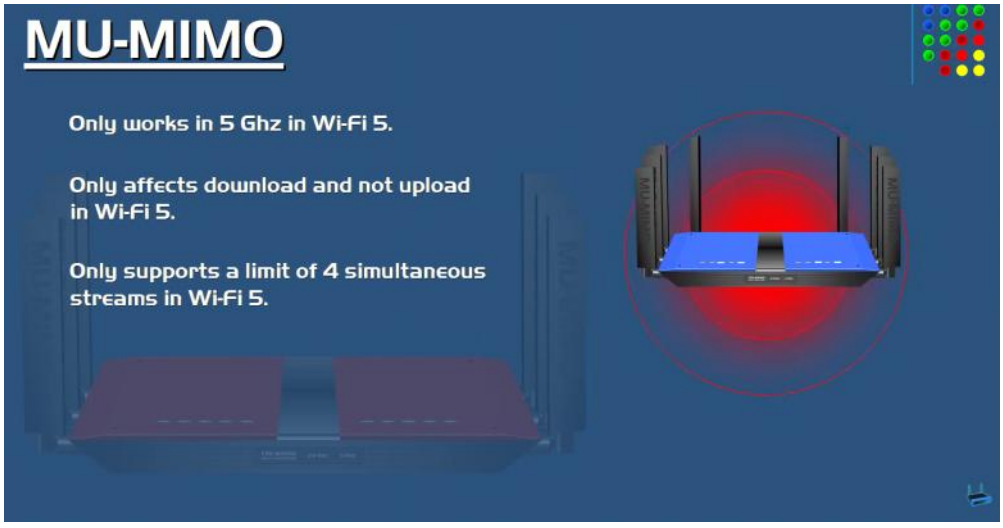


It also uses beamforming to direct signals toward devices, improving connection strength.

For example : this doesn't mean that you cannot add 5th device on your 4x4 Wi-Fi, instead it will share a beam to both the devices but now Wi-Fi 6 can support up to 12 streams

In 802.11ax:

* Works on both 2.4 GHz and 5 GHz
* Supports more streams (8 to 12+)
* Better performance with many devices



Note: Both router and devices must support MU-MIMO, and it mainly improves download performance.